

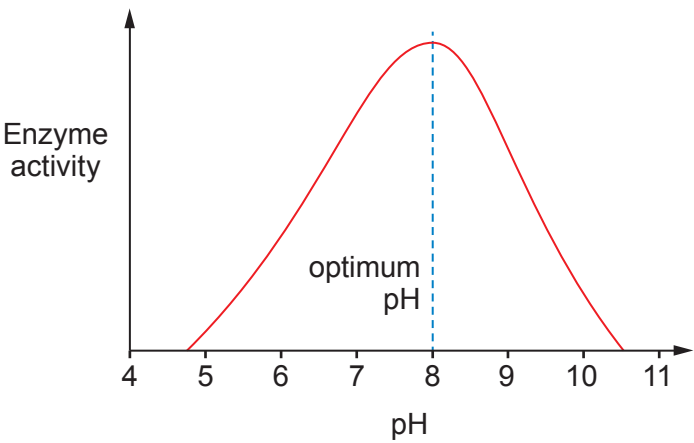
3. This question is about enzymes.

(a) **Complete the following sentence** by underlining the correct word. [1]

Enzymes are made of **carbohydrate / lipid / protein**.

(b) **Graph 3.1** shows the effect of increasing pH on the rate of activity of an enzyme.

Graph 3.1



Use **Graph 3.1** to describe how increasing pH affects enzyme activity. [2]

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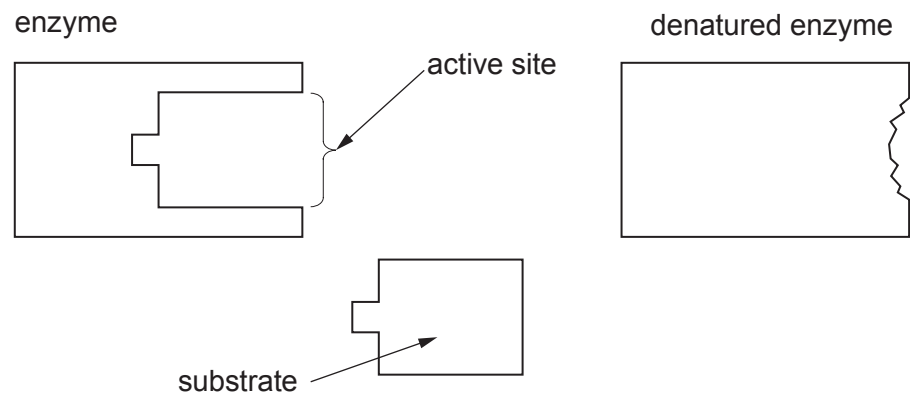
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(c) **Image 3.2** shows an enzyme, a denatured enzyme and a substrate molecule.

Image 3.2



(i) Describe what has happened to the active site in the denatured enzyme. [1]

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(ii) Enzymes react with substrates to produce products.
Explain why the denatured enzyme can no longer produce products. [2]

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(d) State **two** variables that could denature an enzyme. [2]

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